



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

BIOLOGY
PAPER 3 Essays

6030/3
2 hours 30 minutes

NOVEMBER 2024 SESSION

Additional materials:

Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

Answer any **five** questions.

Write your answers on the answer booklet provided.

If you use more than one answer booklet, fasten them together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 3 printed pages.

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- 1 (a) Describe each of the following bonds that hold protein molecules in shape:
- (i) peptide [8]
 - (ii) hydrogen [6]
 - (iii) ionic [6]
 - (iv) disulphide
- (b) Explain the property of water which has a cooling effect in plants and animals. [6]
- (c) Relate the structure of the inner mitochondrial membrane to its function. [6]
- 2 (a) Outline the functions, in the cell membrane, of
- (i) proteins, [4]
 - (ii) phospholipids. [4]
- (b) (i) Compare active transport and facilitated diffusion across cell membranes. [3]
- (ii) Explain the adaptations of cells to active transport. [3]
- (c) Distinguish between protozoa and algae. [6]
- 3 (a) Describe the development of cancer. [8]
- (b) (i) Distinguish between cytokinesis in plant cells and animal cells. [3]
- (ii) Explain the significance of the genetic code being triplet. [3]
- (c) Relate the structure of a nucleotide to its function. [6]
- 4 (a) Outline the functions, in DNA replication, of
- (i) DNA helicase, [1]
 - (ii) DNA polymerase, [3]
 - (iii) DNA ligase, [2]
 - (iv) DNA primase. [2]



- (b) In Mendel's laws of inheritance, explain the law of
- (i) segregation, [3]
 - (ii) independent assortment. [3]
- (c) Explain, using **one** example, the process of artificial selection. [6]
- 5 (a) Describe the observations and deductions made by Charles Darwin in the theory of evolution by natural selection. [8]
- (b) (i) Explain the importance of phosphorylating glucose during glycolysis. [3]
- (ii) Distinguish between cyclic and non-cyclic photophosphorylation. [3]
- (c) Contrast C_3 and C_4 pathways of photosynthesis. [6]
- 6 (a) Describe the structure of a sieve tube. [8]
- (b) Relate to function, the structure of
- (i) an artery, [3]
 - (ii) a vein. [3]
- (c) Explain the stages of the polymerase chain reaction. [6]
- 7 (a) Describe, with reference to gene technology, the role of
- (i) restriction enzymes, [5]
 - (ii) reverse transcriptase. [3]
- (b) Explain how haemoglobin releases oxygen in respiring tissues. [6]
- (c) Explain any **two** factors that affect speed of impulse transmission along neurones. [6]



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

BIOLOGY

PAPER 1 Multiple Choice

NOVEMBER 2024 SESSION

6030/1

1 hour

Additional materials:
Multiple Choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

Read very carefully the instructions on the answer sheet.

Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**, choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet.

INFORMATION FOR CANDIDATES

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

This question paper consists of 13 printed pages.

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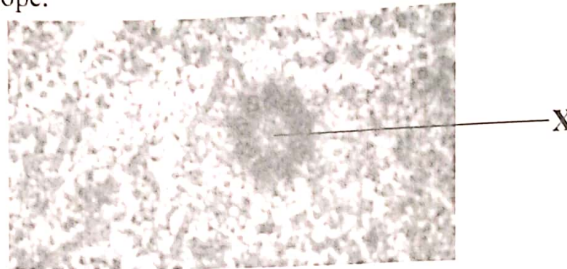


- 1 Which material is used for embedding when preparing material for viewing under a light microscope?
- A ethanol
 - B resin
 - C glutaraldehyde
 - D wax

- 2 Which one does **not** have a membrane?

- A lysosome.
- B nucleolus.
- C phagolysosome.
- D vesicle.

- 3 The electron micrograph shows structures found in a eukaryotic cell as seen under an electron microscope.



What is structure X?

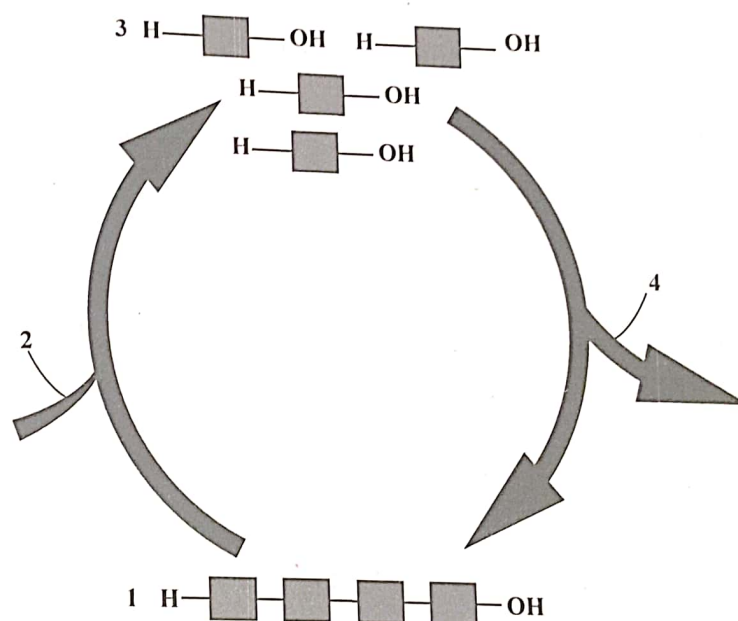
- A centriole
 - B lysosome
 - C nucleolus
 - D ribosome
- 4 What is the thickness of the cell surface membrane?
- A $0.7 \mu\text{m}$
 - B $7.0 \mu\text{m}$
 - C $0.007 \mu\text{m}$
 - D $0.0007 \mu\text{m}$
- 5 Which cell organelle transports synthesized protein around the cell?
- A ribosomes
 - B Golgi apparatus
 - C smooth endoplasmic reticulum
 - D rough endoplasmic reticulum



- 6 Which property of water correctly defines high latent heat of fusion and its significance for life?

	property	significance for life
A	Many substances can dissolve in water.	Medium for the chemical reactions of life.
B	Significant amounts of energy are required before water will change state.	Contents of cells are unlikely to freeze.
C	In order for water to evaporate it must absorb a large amount of energy.	Heat is lost by evaporating of water. Sweating and transpiration causes rapid cooling.
D	Water can absorb a lot of energy for only a small rise in temperature.	Aquatic environments are thermally stable, therefore organisms have stable internal temperatures when the external temperatures is fluctuating.

- 7 The diagram shows the process of polymerisation.



Which row correctly identifies the stages and molecules at 1, 2, 3 and 4?

	1	2	3	4
A	polymer	hydrolysis	monomer	condensation
B	polymer	monomer	hydrolysis	condensation
C	condensation	hydrolysis	polymer	monomer
D	condensation	monomer	hydrolysis	polymer



- 8 Ether is used to extract
- A lipids.
 - B proteins.
 - C carbohydrates.
 - D nucleic acids.
- 9 An enzyme may be inactivated by
- A blocking its active site.
 - B reducing the concentration of the enzyme.
 - C doubling the concentration of the enzyme.
 - D adding more substrates.
- 10 Which one is **not** part of the interphase stage of the cell cycle?
- A replication of DNA
 - B energy stores increase
 - C intensive cellular synthesis
 - D chromosomes become visible
- 11 Which pair correctly represents the pH of histones and DNA?

	histones	DNA
A	acidic	acidic
B	basic	acidic
C	acidic	basic
D	basic	basic

- 12 Which part of the chromosome ensures that no cell death occurs during cell division?
- A centrioles
 - B centromeres
 - C centrosomes
 - D telomeres
- 13 The following statements show the significance of nuclear division.
1. Promotes genetic variation
 2. Asexual reproduction occurs
 3. Prevents doubling of chromosomes

Which statement(s) show the significance of meiosis?

- A 1 and 2
- B 1 and 3
- C 2 and 3
- D 2 only



- 14 The numbered events occur during different phases of meiosis.

1. Bivalents arrange on the equator.
2. Spindle fibres contract and pull the homologous chromosomes.
3. Spindle fibres disappear and chromatids recoil.
4. Nucleolus disappears and nuclear envelope breakdown.

What is the correct sequence of the events?

- A 4 → 1 → 2 → 3
 B 2 → 1 → 3 → 4
 C 1 → 4 → 2 → 3
 D 3 → 1 → 2 → 4

- 15 A purine base contains five carbon atoms and a pyrimidine contains four carbon atoms.

What is the number of carbon atoms in a nucleotide containing the base guanine?

- A 4
 B 5
 C 9
 D 10

- 16 On a DNA molecule, the codon TAC codes for methionine.

What is the anticodon on the tRNA to which methionine becomes attached?

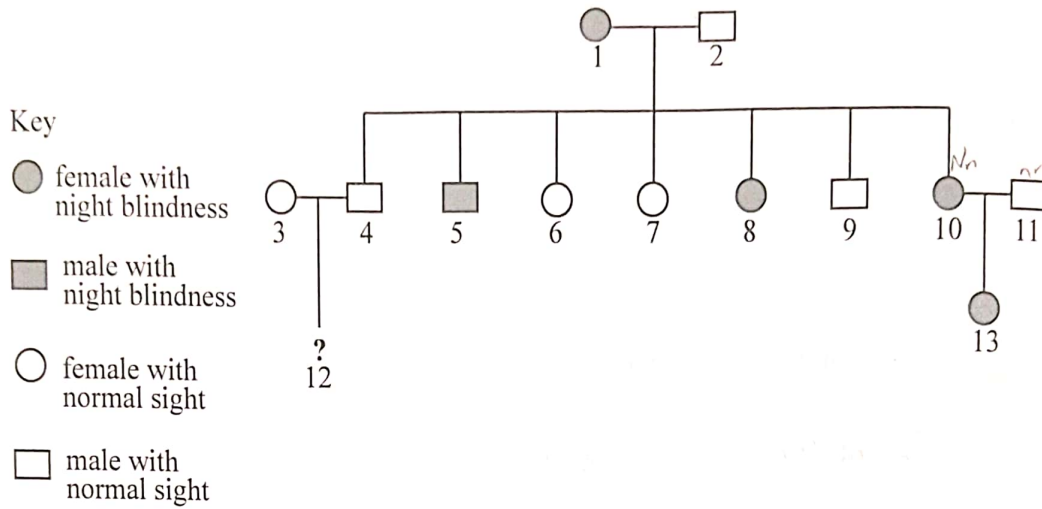
- A AGU
 B AUG
 C UAC
 D UTG

- 17 Which tRNA base sequences corresponds to the DNA base sequence TGACCGTTA?

- A ACT GGC AAT
 B UGA CCG UUA
 C UGA GGC AAU
 D ACU GGC AAU



- 18 Night blindness is a condition in which affected people have difficulty seeing in dim light. The allele for night blindness, N , is dominant to the allele for normal vision, n . (these alleles are not on the sex chromosomes). The diagram below represents part of a family tree showing the inheritance of night blindness.



What is the probability that the next child born to individuals 10 and 11 will be a girl with night blindness?

- A 0
 - B 0.25
 - C 0.50
 - D 1
- 19 Which one is **not** a stage of the Calvin Cycle?
- A regeneration of RuBP
 - B reduction of NADP
 - C carbon dioxide fixation
 - D glycerate phosphate reduction
- 20 Substrate level phosphorylation takes place during both
- A glycolysis and the Krebs Cycle.
 - B the electron transport system and the link reaction.
 - C glycolysis and the electron transport system.
 - D the link reaction and the Krebs Cycle.



- 21 In an experiment, six tubes were set up as shown in the table.

tube	content
1	glucose + homogenised animal cells
2	glucose + mitochondria
3	glucose + cytoplasm lacking organelles
4	pyruvate + homogenised animal cells
5	pyruvate + mitochondria
6	pyruvate + cytoplasm lacking organelles

After incubation, each sample was analysed to determine the presence of carbon dioxide and lactate.

In which tubes is lactate most likely to be present?

- A 1 and 3 only
- B 2, 3, 5 and 6
- C 3 and 6 only
- D 4, 5 and 6 only.

- 22 The xylem vessel is lignified to

- A help adhesion of water molecules.
- B allow lateral flow of water.
- C increase water pressure.
- D to get rid of air bubbles.

- 23 Cells involved in transport have the following features.

- ✓ 1. lignified wall
- 2. cytoplasm
- 3. mitochondria
- ✓ 4. pits
- 5. plasmodesmata

Which of the features are present in tracheids?

- A 1, 2 and 3
- B 1, 4 and 5
- C 2, 4 and 5
- D 2, 3 and 5

- 24 Which of the following diseases is highly prevalent in developed countries?

- A coronary heart disease
- B HIV and AIDS
- C sickle cell anemia
- D tuberculosis

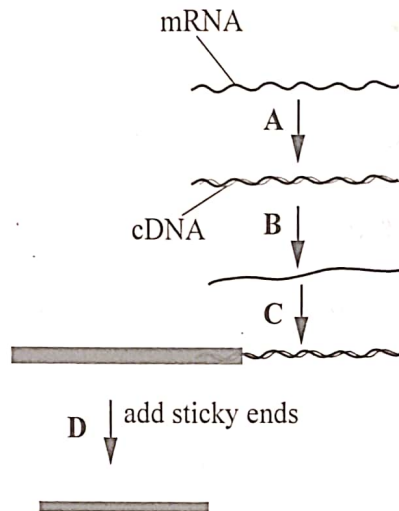


25 Which disease can spread rapidly across international borders?

- A CHD
- B malaria
- C HIV/AIDS
- D cholera

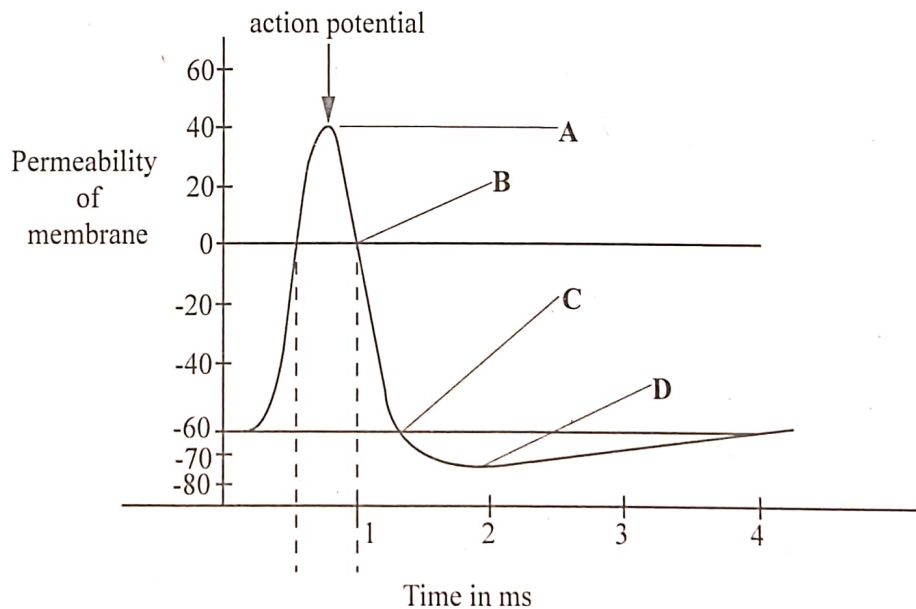
26 The diagram shows how the insulin gene can be obtained from mRNA.

During which stage is *reverse transcriptase* used?



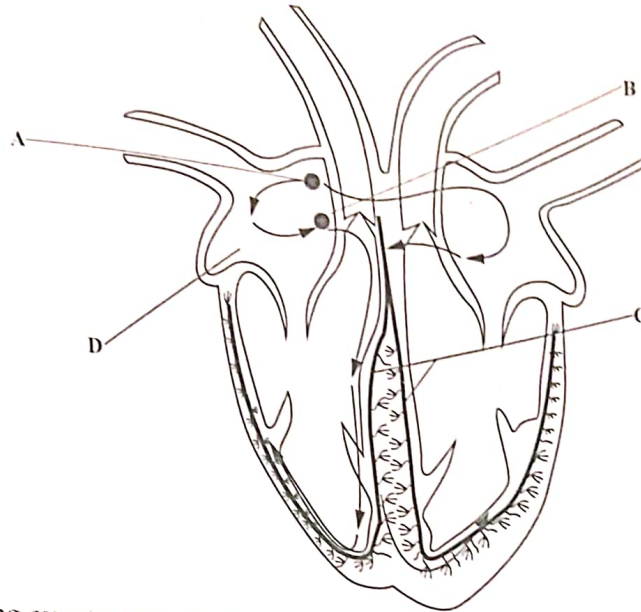
27 The graph shows the action potential across an axon membrane.

At which point, A, B, C and D do the potassium gates close during the action potential?



28 The diagram shows parts of the heart.

Which part, A, B, C or D is the pacemaker?



29 The following events occur during the cardiac cycle.

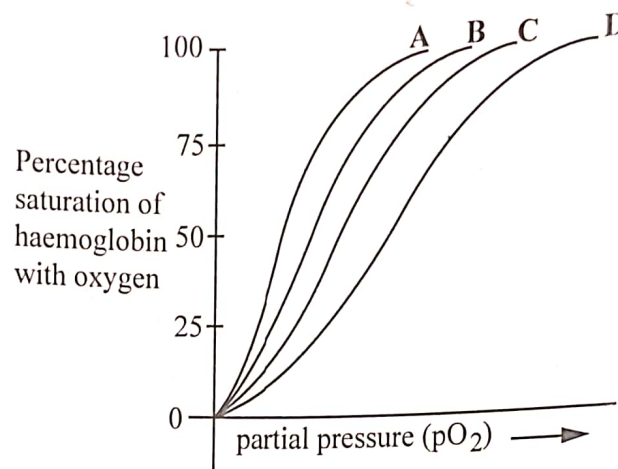
- 1 impulse spreads across atria
- 2 SAN excited
- 3 AVN delays impulse
- 4 Atria contract
- 5 Ventricles contract
- 6 Impulse passes through bundle of His
- 7 Impulse moves from base of ventricles

What is the correct order of the events?

- A 2 → 1 → 3 → 4 → 6 → 5 → 7
- B 2 → 1 → 4 → 3 → 6 → 7 → 5
- C 1 → 2 → 4 → 3 → 7 → 6 → 5
- D 1 → 2 → 3 → 6 → 5 → 4 → 7

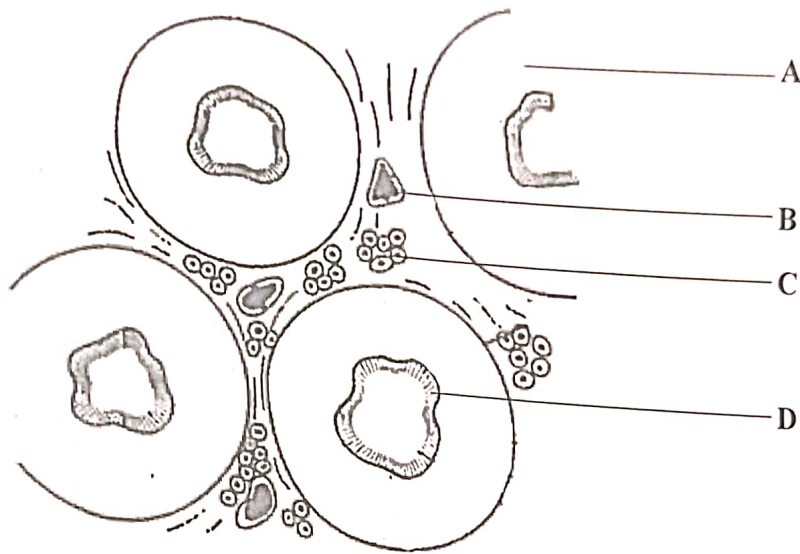


- 30 The graph shows the oxygen dissociation curves of four mammals. Which curve is for a mammal living at the lowest altitude?



- 31 At which stage of an action potential does the axon membrane experience positive feedback effect?
- A during the depolarisation stage
 - B during the hyperpolarisation stage
 - C during the repolarisation stage
 - D during the resting stage
- 32 Which one is a “glandular” function of the placenta?
- A Exchange of materials between the mother and the foetus.
 - B Protecting the foetus by acting as a barrier to pathogens.
 - C Production of lactogen to stimulate the enlargement of mammary glands.
 - D Providing passive immunity by allowing passage of antibodies from the mother to the foetus.

- 33 The diagram represents a group of seminiferous tubules.
In which part does spermatogenesis occur?

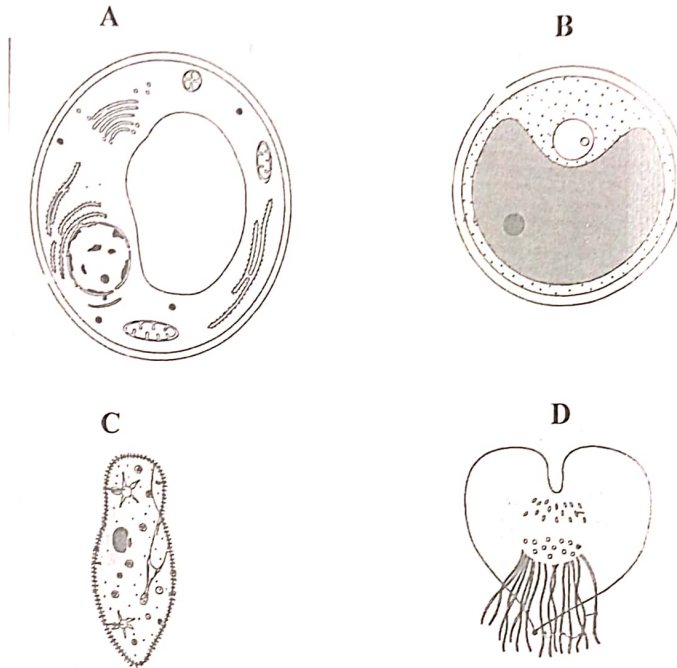


- 34 Which of the following hormones is inhibited by oestrogen?
- A progesterone
 - B leutinising hormone
 - C follicle Stimulating Hormone
 - D gonadotrophic releasing hormone
- 35 How many divisions does the microspore nucleus undergo before fertilisation?
- A 1
 - B 2
 - C 3
 - D 4
- 36 A chemical that kills *Nitrobacter* was used in an Agricultural system.
What is the effect of this chemical on the nitrogen cycle?
- A ammonia production increases
 - B ammonium production increases
 - C nitrite production increases
 - D nitrate production increases

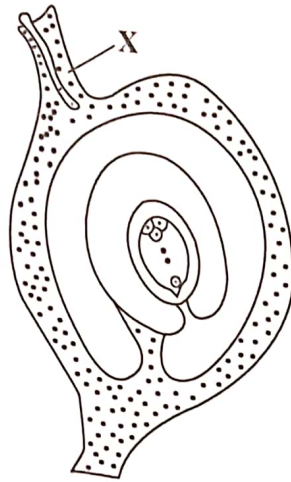
37 What is the chemical indicator of eutrophication?

- A Biochemical nitrogen demand.
- B Biological oxygen demand.
- C Biochemical oxygen demand.
- D Biodiversity oxygen demand.

38 Which organism belongs to the kingdom Fungi?



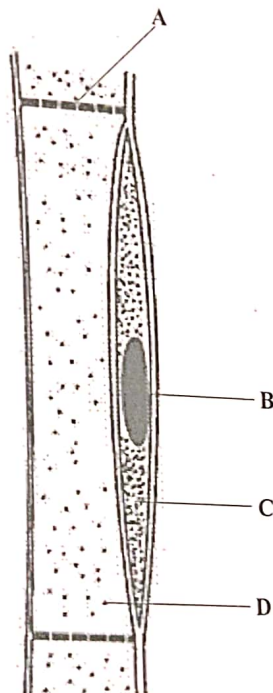
- 39 The diagram shows a mature embryo sac inside a flower.



What is structure X?

- A stalk
- B integument
- C style
- D ovary wall

- 40 The diagram represents a sieve tube element and its companion cell.
- Which part prevents the sieve tube from budding?



Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

BIOLOGY

PAPER 4 Practical Test

6030/4

2 hours 30 minutes

NOVEMBER 2024 SESSION

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
TOTAL	

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Question 1

You are required to investigate the effect of pH on the action of the enzyme chymosin on milk.

You are provided with fresh cow's milk, enzyme chymosin and buffered solutions at pH 1.5, 2.0, 2.5, 3.0 and 3.5.

Proceed as follows:

Label 5 petri-dishes: pH1.5, pH2.0, pH2.5, pH3.0 and pH3.5.

Use a clean 10 cm³ syringe to pipette 10 cm³ of cow's milk into each petri-dish.

To petri dish pH1.5, add 2 cm³ of enzyme chymosin using a clean 5 cm³ syringe.

To the same petri dish, pH1.5, add 5 cm³ of the buffer solution at pH1.5. Use a clean syringe.

Immediately start the stop watch and observe the contents of the petri dish for any signs of curdling.

When the first signs of curdling are observed stop the watch, and record the time in Table 1.1.

(a) Table 1.1

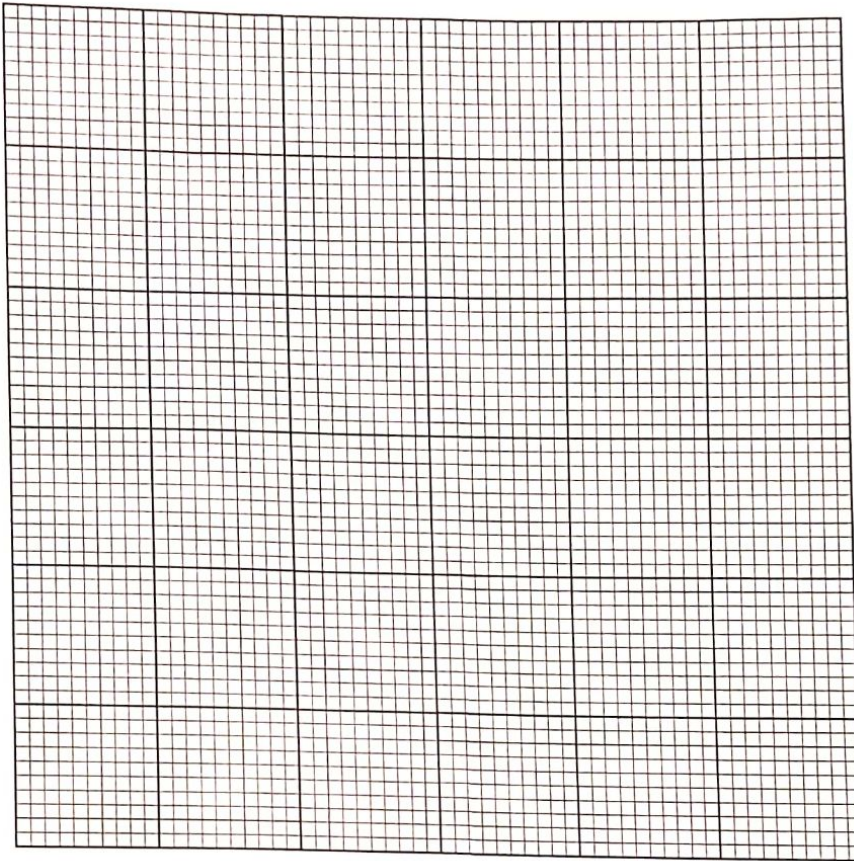
pH	time taken to curdle milk (sec)
1.5	
2.0	
2.5	
3.0	
3.5	

[5]

Repeat the procedure using the remaining four petri dishes, taking care to add the enzyme first before adding the appropriate buffer solution.



- (b) Plot an appropriate graph using the results in Table 1.1.



[4]

- (c) (i) Explain the shape of the curve.

[3]

- (ii) Use the graph to estimate the optimum pH of chymosin.

[1]



- (d) (i) Explain how chymosin is causing curdling in milk.

_____ [2]

- (ii) State **one** commercial use of chymosin.

_____ [1]

- (e) Suggest ways of improving the reliability of the results in this investigating.

_____ [2]

[Total: 18]



Question 2

You are required to investigate the effect of alcohol on the amount of pigment leaking through beetroot plasma membranes.

Proceed as follows:

Label six (6) test tubes **A, B, C, D, E** and **F**.

Use the 100% extract provided to prepare equal volumes of standards ranging from 0 – 100%.
NB: Prepare the standards in test tubes **A** to **F** respectively.

- (a) Complete **Table 2.1** to show how the standards will be made.

Table 2.1

	A	B	C	D	E	F
Volume of beetroot Extract/cm ³				3		
Volume of distilled Water/cm ³	5					
Concentration of extract/%	0					100

[3]

Prepare the standards and place them in a test tube rack.

Label five (5) boiling tubes **G, H, I, J** & **K**.

Transfer, 5 cm³ of 100% alcohol into tube **G**,
 5 cm³ of 50% alcohol into tube **H**,
 5 cm³ of 25% alcohol into tube **I**,
 5 cm³ of 10% alcohol into tube **J**,
and 5 cm³ of 5% alcohol into tube **K**.

Immediately put a bung onto each tube.

Place tube **G – K** in a water bath maintained at 30°C for 5 minutes.

After 5 minutes, blot 10 discs of beetroot tissue with paper towel to remove excess pigment.

Transfer two discs into each of the 5 tubes **G – K**, and replace the bungs.

Leave the tubes in the water bath for 5 minutes, shaking them gently once every minute.

After 5 minutes, tilt each boiling tube once, and decant each solution into a clean test tube. Be careful to label the test tubes appropriately.

Compare the collected solutions with the standards set aside on the test tube rack.

Estimate the pigment concentration for each collected sample, using the standards **A – F**.



- (b) Record the estimates in a suitable table in the space provided.

[4]

- (c) Explain the trend of the estimates in the table in (b).

[4]

- (d) Explain why the beetroot had to be washed prior to the experiment.

[1]

- (e) Suggest how the experiment could be improved to obtain more reliable results.

[2]



(f) Predict, with a reason, the expected observation if

(i) the beetroot had been cooked.

[2]

(ii) the water bath was maintained at 10°C.

[2]

(iii) a detergent, instead of alcohol was used.

[2]

[Total: 20]



Question 3

N24 is a transverse section of a blood vessel from a mammal.

Examine **N24** carefully under low power of the microscope.

- (a) Make a plan drawing to show the distribution of tissues that make **N24**.

[4]

- (b) You have been provided with an eyepiece graticule fitted to the microscope and a stage micrometer.

Calibrate the eyepiece graticule under low power of the microscope using the stage micrometer.

- (i) Calculate the size of one eyepiece unit. Show working.

Eyepiece unit = _____ μm

[3]



- (ii) Measure the diameter of N24 using the eyepiece graticule.
Show any working.

Diameter = _____ [2]

Draw a line, X, across the diameter of the plan drawing in (a).

- (iii) Measure and record the length of line X drawn in the plan drawing.

X = _____ mm

Calculate the magnification of the lumen in the plan drawing. Show Working.

Magnification = _____

[3]
[Total: 12]



To be provided by centre

Each candidate must be provided with the following apparatus and materials.

To be provided by the centre

1. 100 cm³ 10% yeast suspension.
2. Four boiling tubes.
3. Boiling tube rack.
4. 10 cm³ 1% methylene blue labelled as such.
5. Dropper pipette.
6. 20 cm³ 2% glucose solution labelled as such.
7. Four 600 cm³ beakers / large container to use as water bath
8. Bunsen burner
9. Access to cold water (Below 10°C)
10. Means of making glassware
11. Stop watch
12. Thermometer (-10°C - 110°C)
13. 5 × 10 cm³ syringes
14. stirring rod × 1
15. syringes: 2 cm³ × 6
16. teat pipette × 1
17. Benedict's solution (10 cm³)
18. Beaker 400 ml
19. Test tubes × 7
20. Copper Sulphate (1%) – 5 cm³
21. Sodium hydroxide (5%) – 10 cm³
22. Stop watch
23. Extracts E (Bean), F (Matemba / dried kapenta) and G (Soya chunks) – 10 cm³ each.
24. Access to a burner.
25. 3 cm³ of plain yoghurt labelled **X**
26. a blank microscope slide
27. a coverslip
28. a glass rod
29. a microscope with × 10 and × 40 objective lenses

The supervisor should soak foodstuffs overnight / 24 hrs. before the exam, freshly prepare the extracts by crushing / grinding / then filter through layers of mutton cloth. finally filter paper to obtain clear extracts.

Crushing **10g** of a sample in 50 cm³ of distilled water gives enough extract for three candidates. Extracts to be labelled **E, F and G**.

Candidate Name _____

Centre Number _____

Candidate Number _____



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

BIOLOGY

PAPER 2 Theory Structured

NOVEMBER 2024 SESSION

6030/2

1 hour 30 minutes

Additional materials:
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
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9	
10	
TOTAL	

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Answer all questions

- 1 (a) Fig. 1.1 shows the general structure of a plant cell.

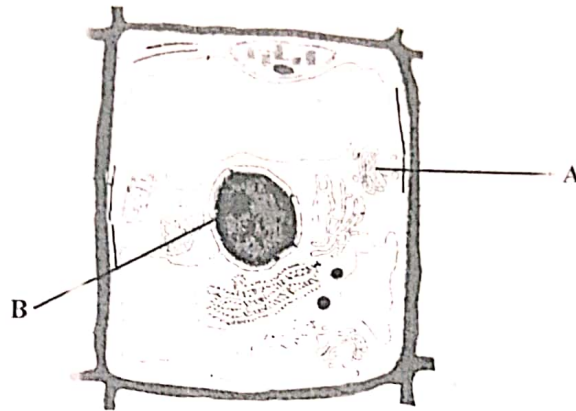


Fig. 1.1

- (i) Identify the part labelled A.

[1]

- (ii) Describe the function of part B.

[1]

- (b) (i) Contrast between magnification and resolution.

[1]

- (ii) Illustrate how the magnification is determined in a light microscope with $\times 10$ objective lens and a $\times 5$ eye piece lens. Show working.

[2]

[Total: 5]



- 2 (a) Fig. 2.1 shows an organelle used during translation.

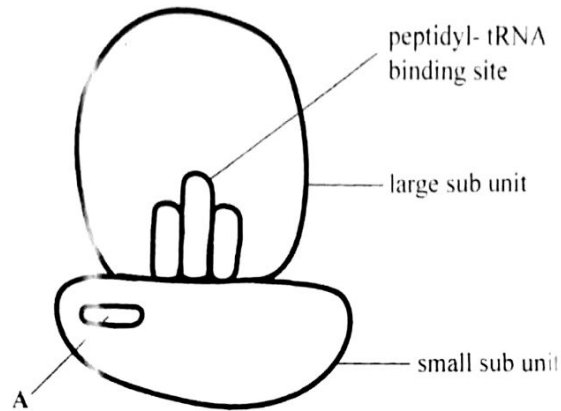


Fig. 2.1

- (i) Name the nucleic acid which binds to site A.

_____ [1]

- (ii) Identify the organelle in Fig. 2.1.

_____ [1]

- (iii) Predict the effect of inhibiting peptidyl tRNA transferase in protein synthesis.

 _____ [1]

- (b) Explain post-translation modification of proteins.

 _____ [3]
 [Total: 6]

- 3 (a) Fig. 3.1 shows a stage during pregnancy.

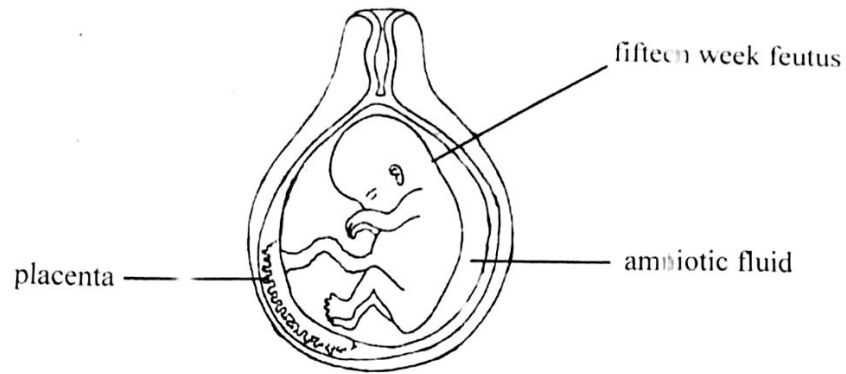


Fig. 3.1

- (i) Identify a technique used to extract a DNA sample.

[1]

- (ii) Describe the procedure for the technique identified in (a)(i).

[2]

- (b) Suggest at least **three** reasons why genetic counselling is necessary before carrying genetic screening.

[3]

[Total: 6]

CHC

5

use 1

BA

Fig. 4.1 shows positions of photosynthetic pigments on a chromatogram.

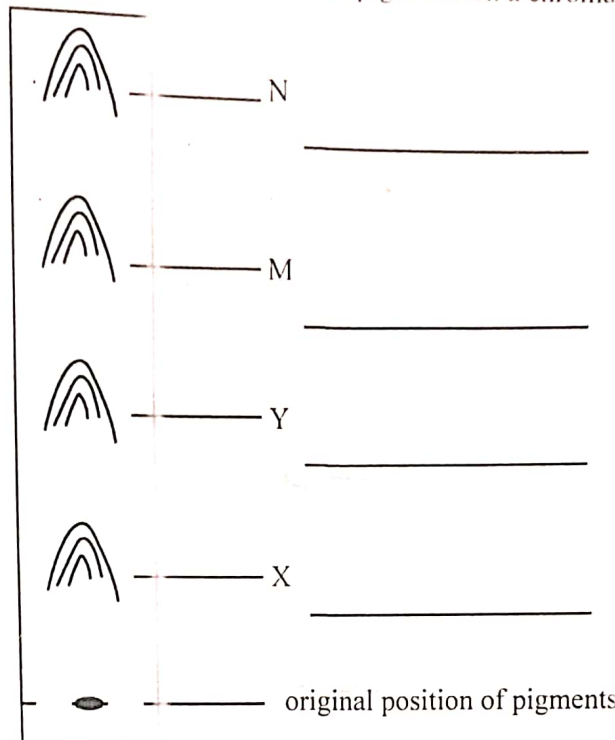


Fig. 4.1

- (a) State any **two** factors affecting the separation of the pigments.

- (b) On the spaces provided on Fig. 4.1, match the following pigments with the positions on the chromatogram. Carotene, chlorophyll *a*; xanthophyll; and chlorophyll *b*.

[2]

[4]

- (c) Suggest any **one** use of pigments in life, other than in photosynthesis.

[1]

[Total: 7]

For
Examiner's
Use



Fig. 5.1 shows the structure of a capillary bed.

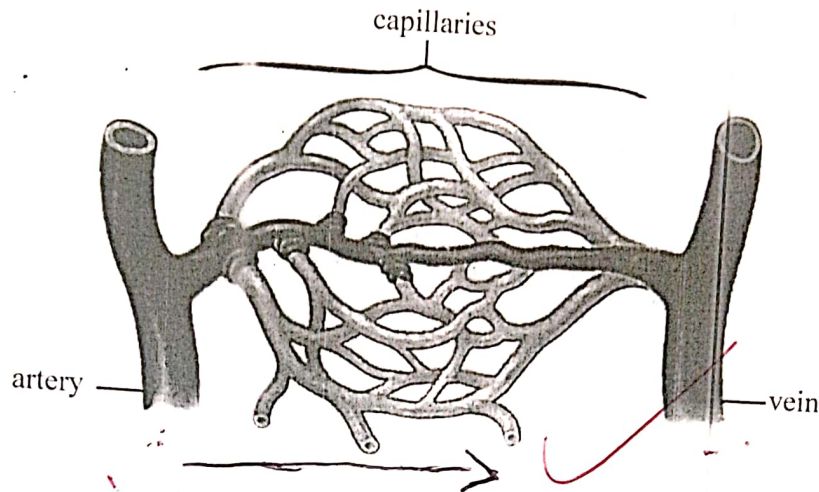


Fig. 5.1

- (a) (i) On Fig. 5.1, indicate, with an arrow, the direction of blood flow. [1]

- (ii) State the main driving force responsible for blood to flow in the direction indicated in (a)(i).

Blood pressure generated by the contraction of the heart. [1]

- (iii) In ascending order, arrange the blood vessel in Fig. 5.1 according to the velocity of blood flow.

vein → capillaries → artery
artery, arterioles, capillary, venules, vein [1]

- (b) Relate the structure of capillaries to their function.

Composed of a single layer of endothelial cells, allowing for rapid diffusion of substances across the capillary wall. The network of capillaries provides a large surface area for exchange, allowing for efficient transfer of oxygen, nutrients and waste products.

Narrow lumen:

[3]

[Total: 6]

Large lumen relative to diameter as blood is flowing under low pressure allowing for exchange of materials & tissue.



Vib.

6

(a) Define the term *resting potential*.

 _____ [1]

(b) Fig. 6.1 shows part of an axon membrane.

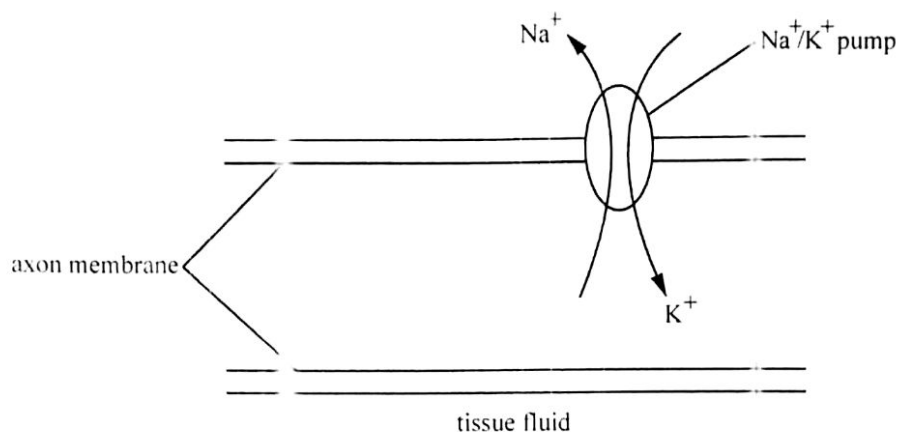


Fig. 6.1

(i) Describe the concentration of sodium ions and potassium ions in the tissue fluid.

 _____ [2]

(ii) Suggest how the sodium potassium pump maintains the resting potential across the axon membrane.

 _____ [2]

(iii) Explain how an increase in the axon diameter can affect nerve impulse transmission along the axon.

 _____ [1]
 [Total: 6]



7 Fig.7.1 is an illustration of hormonal control during spermatogenesis.

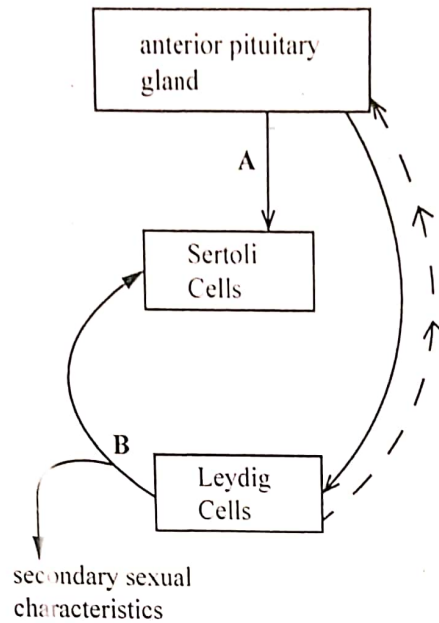


Fig. 7.1

(a) Name the hormone

A. _____

B. _____

[2]

(b) State the process indicated by the dotted arrow.

_____ [1]

(c) A male contraceptive pill contains testosterone.

(i) Explain the effect of the pill on gametogenesis.

[2]



- (ii) Suggest ethical implications associated with male contraceptive pills.

For
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Use

[2]

[Total: 7]

- 8 (a) Table 8.1 shows information of some terms that are used in ecology.

Complete Table 8.1 by filling in the missing information.

Table 8.1

term	definition
habitat	A place where an organism lives.
	All the organisms and the physical factors that influence them in an area.
community	

[2]

- (b) Fig. 8.1 illustrates one of the processes in the nitrogen cycle.

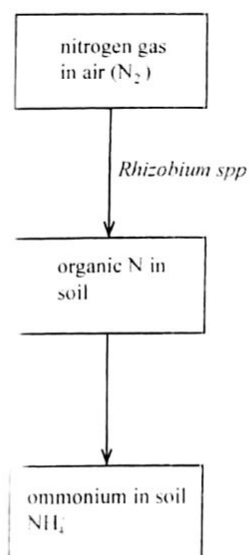


Fig. 8.1

- (i) Name the process

[1]

6030/2 N2024

[Turn over]

- (ii) *Rhizobium* spp is aided by nitrogenase enzyme to produce NH_4^+ .

Explain any other **two** conditions required by the bacteria in carrying out the process.

condition _____

explanation _____

condition ? _____

explanation _____

[4]

[Total: 7]

- 9 Classification involves placing living organisms in a series of increasingly specialised groups based on similarities and differences.

- (a) State the name given to identify the science of classification.

[1]

- (b) Name any **one** branch of classification.

[1]

- (c) Explain why scientists prefer using phenetic classification to phylogenetic classification.

[3]

[Total: 5]

Fig. 10.1 shows estimated projections between incidence rate and death rates for T.B.

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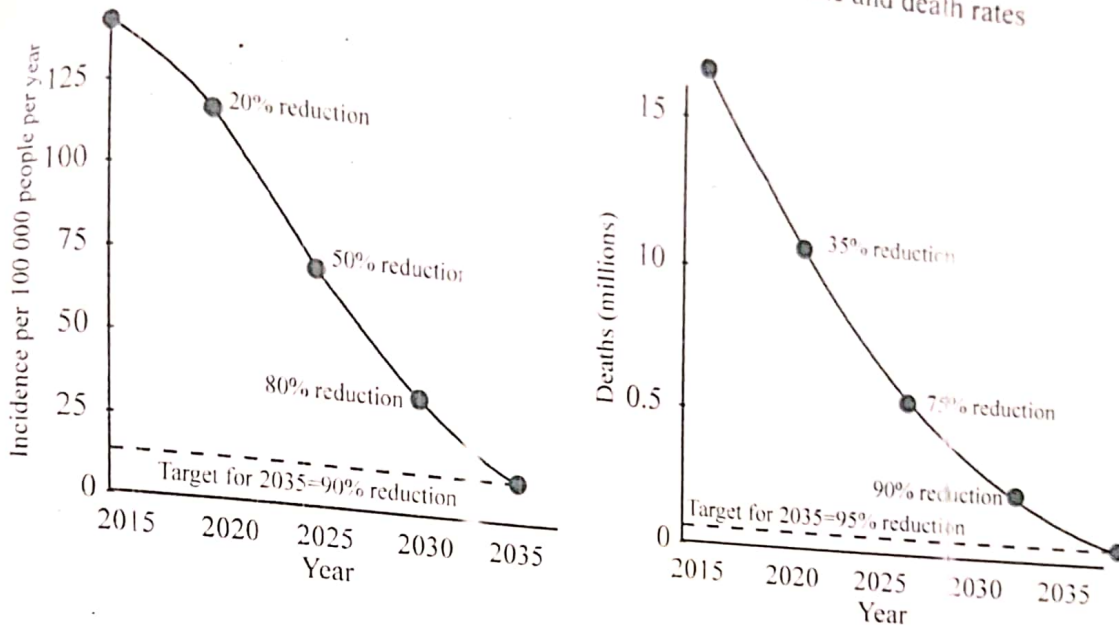


Fig. 10.1

- (a) Define the terms:

endemic _____

[1]

incidence _____

[1]

- (b) Describe the relationship between incidence rate and mortality rate in Fig. 10.1.

[2]

- (c) Suggest a possible way to achieve the projected reduction in death rates by 2035.

[1]

[Total: 5]